Program : **Diploma in Chemical Engineering**	
Course Code : **4078**	Course Title: **Fluid Mechanics Lab (Chemical)**
Semester : **4**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- To provide hands-on experience in the working of fluid handling equipment and measuring devices.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Properties of matter		Engineering Physics-I	1
Unit operations in chemical engineering		Chemical Process Principles	2
Properties of fluids, Fluid friction and flow measurement and Fluid handling equipment		Fluid Mechanics	3

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Compute the velocity profile and frictional losses in pipes.	6	Applying
CO2	Interpret flow metering devices.	18	Applying
CO3	Evaluate the characteristics of a centrifugal pump.	12	Applying
CO 4	Illustrate the effect of temperature on viscosity and verify Stock's law.	9	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2		3					
CO3		3					
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Compute the velocity profile and frictional losses in pipes.**		
M1.01	Interpret the flow pattern in horizontal pipe and relate them with Reynolds Number.	3	Applying
M1.02	Compute the friction factor in pipes and fittings for different flow rate.	3	Applying
CO2	**Interpret flow metering devices.**		
M2.01	Determine the coefficient of discharge of a Ventury meter.	3	Applying
M2.02	Determine the coefficient of discharge of an Orifice meter.	3	Applying
M2.03	Calibrate the given Rotameter and hence estimate the flow rate.	3	Applying
M2.04	Determine the coefficient of discharge of a notch.	3	Applying
M2.05	Plot the velocity profile for air flow through a circular pipe using pitot tube.	3	Applying
M2.06	Open ended experiments	3	Applying
	Series Test – I		
CO3	**Evaluate the characteristics of a centrifugal pump.**		
M3.01	Demonstrate the working of a centrifugal pump.	3	Applying
M3.02	Plot the characteristic curves of the given Centrifugal pump and compare with the manufacturers data.	3	Applying
M3.03	Estimate the maximum efficiency of the given Centrifugal pump.	3	Applying
M3.04	Open ended experiments	3	Applying

CO4	**Illustrate the effect of temperature on viscosity and verify Stock's law.**		
M4.01	Effect of temperature on viscosity using Redwood viscometer	3	Applying
M4.02	Verify stokes law	3	Applying
M4.03	Open ended experiments	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Unit Operations; Warren.L.McCabe Julian.C.Smith
R1	Introduction to chemical Engineering; Walter.L.Badger & Julius.T.Banchero
R2	Fluid Mechanics; Yunus A Cengel, John M Cimbala

Online Resources

Sl.No	Website Link
1	https://www.vlab.co.in/broad-area-chemical-engineering